

COSC 519 Final Project Report

Courtyard Reflections: An Immersive Journey Through Academic Stress at UBC Okanagan

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1 Project Overview

Courtyard Reflections is a Virtual Reality (VR) experience designed to make the invisible escalation of academic stress visible, audible, and felt. The player inhabits a UBCO student whose ordinary day gradually transforms from manageable pressure into cognitive overload, before resolving into reflection, grounding, and mental health awareness.

The experience is implemented in Unity using URP, XR Interaction Toolkit, and UI Toolkit. Its modular systems include dialogue management, task scheduling, notification overflow, stress visualisation, and adaptive audio. The narrative follows Kishōtenketsu (Ki-Shō-Ten-Ketsu), a non-adversarial structure that enables emotional escalation without combat or failure mechanics.

1.1 Motivation

Rising academic pressure and cognitive overload are increasingly common among university students, yet much of this experience remains internal and difficult to articulate. Early signs of stress are often overlooked or normalised until they accumulate into a sense of overload. While written resources describe these challenges, they do not always convey the emotional progression that students experience in real time.

Immersive technologies offer an opportunity to externalise these internal processes. Virtual reality, specifically, can represent cognitive strain through narrative pacing, environmental modulation, and embodied perspective, allowing users to recognise stress patterns in ways that traditional media cannot. Hence, we use VR to depict the buildup of academic stress, encourage self-reflection, and connect students to appropriate wellbeing resources. By grounding the experience in familiar settings at UBC Okanagan, we try to spread awareness, reduce stigma around help-seeking, and contribute to broader conversations about student mental health.

2 Literature-Informed Narrative & Experience Design

2.1 Narrative as Emotional Trajectory

The experience is structured as an emotional curve rather than a mission-based progression. Murray [13] frames digital narrative as the shaping of “cyberdrama,” where emotional engagement over time is the primary metric. Laurel [10] similarly positions interactive systems as theatrical structures that guide users through tension and release.

Each act in our design combines:

- a social interaction (friend, professor, inner monologue),
- a task-management decision,
- changes in environmental and bodily feedback (soundscape, post-processing, heartbeat).

Additionally, Mateas and Stern’s interactive drama framework [11] emphasises guided agency, where choices adjust tone but not the underlying emotional arc. So, our design

follows this principle: choices personalise the pathway to overload without removing overload itself, enabling comparative analysis across users.

2.2 Kishōtenketsu as a Non-Adversarial Stress Structure

Kishōtenketsu provides a non-conflict-based structure suitable for mental health content.

- **Ki**: Introduce courtyard, friend, and manageable stressor.
- **Shō**: Responsibilities accumulate (TA workload).
- **Ten**: Internal overload becomes the twist; the “antagonist” is cognitive collapse.
- **Ketsu**: Integrate experience into awareness and resource-seeking.

This avoids framing stress as a “battle” and instead represents it as a natural trajectory.

2.3 Presence, Environmental Feedback, and Stress Externalisation

Slater and Wilbur’s FIVE framework [14] positions immersion and presence as foundational to impactful VR. McMahan [12] highlights that environmental reactivity strengthens psychological realism.

Environmental feedback operationalises stress as:

- ambient audio modulations (chatter, wind, distortion),
- post-processing (vignette, grain, chromatic aberration),
- heartbeat audio tied to sanity-to-stress mapping.

This continuous mapping externalises internal stress without relying solely on dialogue.

2.4 Abstract NPCs and Colour-Coded Emotion

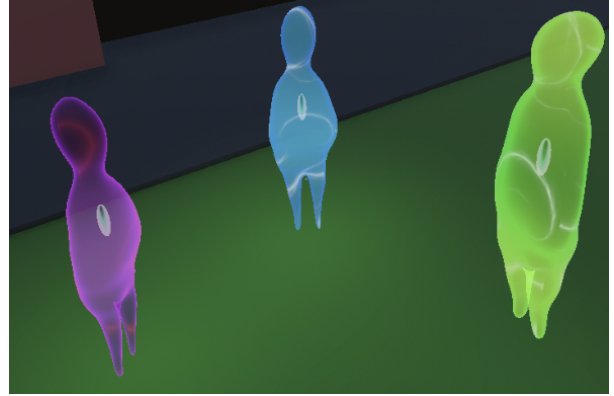
NPCs are deliberately faceless, stylized silhouettes with emissive colour gradients rather than human-like faces. This decision combines three strands of prior work:

Colour as affective shorthand: Valdez and Mehrabian show that colour hue, saturation, and brightness systematically modulate emotional responses (e.g., red associated with high arousal, blue with calm) [16]. Elliot and Maier’s color-in-context theory further argues that colour can shape affect, cognition, and behaviour depending on context [6]. We leverage this by assigning:

- Warm reds and oranges to interactable, stressed NPCs (Figure 1a)
- cooler greens, blues, and purples to background peers in varying emotional states. (Figure 1b)



(a) Interactable NPC with warm red/orange gradient



(b) Background NPCs with cool color gradients

Figure 1: Colour-coded NPC design: Interactable NPCs use warm colours to signal stress and importance, while background characters use cooler tones to indicate peripheral emotional states.

This lets players read the “emotional temperature” of each figure at a glance, with very low cognitive cost, which matters because the user is already under simulated stress.

Abstraction to support projection and empathy. Research in psychology and HCI consistently demonstrates that lower-fidelity or abstract characters increase users’ ability to project their own emotional interpretations onto virtual agents. Studies on character stylization show that when visual detail is reduced, particularly in facial features, users engage more deeply in interpretative empathy, filling in emotional meaning from personal experience rather than relying on predefined facial cues.

For example, Lankoski and Järvinen [9] argue that abstract characters support stronger emotional engagement because they “leave conceptual space” for the player’s imagination, allowing self-referential interpretation common in role-play environments. Similarly, Takatalo et al. [15] demonstrate that simplified avatars can enhance affective alignment and reduce cognitive load, producing clearer emotional resonance. Bergström et al. [2] found that stylized or non-photorealistic agents often evoke greater social comfort and empathy compared to highly detailed human-like avatars, particularly when the scenario involves sensitive emotional states.

These works collectively argue that abstraction strengthens identification and emotional projection, which is essential for a mental-health-oriented narrative where the user must interpret stress cues without being overwhelmed by detailed anthropomorphic expression.

Overall, the NPC design trades photorealism for emotional clarity and gives users interpretive space: they know that “this figure is stressed or calm” without the system dictating

exact facial expressions.

3 Narrative and Interaction

This section explains each act with explicit emphasis on how the storyline, environmental feedback, audio, and interaction are coordinated to shape the user’s emotional journey, based on the theoretical framing.

3.1 Act 0: Title & Onboarding

3.1.1 Narrative Goal

Introduce the user to the courtyard environment, establish a grounded starting point, and emphasize comfort, control, and safety before emotional escalation begins. This primes the user for relational grounding and highlights how stress later disrupts normal routines.

3.1.2 Story and Core Interaction

- A calm introductory voiceover welcomes the user to the UBCO courtyard and explains that they are stepping into a typical student day.
- The user is reminded of their agency and ability to pause or exit at any time.
- The first NPC (a friend) appears with a glowing yellow interaction ring, inviting the user to a small evening event.
- The player chooses between two responses:
 - **Excited:** “Yeah, that sounds nice. Count me in.”
 - **Unsure but willing:** “I’m not totally sure about my evening yet, but I’ll try to make it.”
- Both choices frame the day as manageable and socially supported.

These expressive choices do not branch the story but let users anchor their emotional stance early, consistent with the concept of expressive agency [13].

3.1.3 Environmental Feedback

Ambient audio consists of low chatter and light wind. The courtyard is well-lit with neutral tones. There are no post-processing effects. This supports comfort and gradual entry, reflecting Laurel’s emphasis on designing for embodied user state [10].

3.2 Act 1: The First Crack (Ki: Introduction)

3.2.1 Narrative Goal

Introduce the first academic stressor through a social interaction. This establishes how external responsibilities begin to accumulate and disrupt the user’s sense of control.

3.2.2 Story and Core Interaction

- After completing the first interaction, the same friend returns, now sounding more stressed
- The friend informs the user that a lab report deadline has been suddenly moved up to tomorrow.
- Player responses:
 - **Overwhelm:** “Tomorrow? I’m already behind. I don’t know if I can keep up.” *Friend:* “Honestly, same. Everyone’s barely holding it together this week.”
 - **Avoidance:** “I can’t deal with that right now. I’ll think about it later.” *Friend:* “Yeah, sometimes ignoring it feels easier than facing it. This semester is relentless.”
- The friend empathizes with either emotion, reinforcing the shared stress among students.
- The conversation ends with the friend leaving, signaling rising collective pressure.

3.2.3 Environmental Feedback

- **Phone Interaction:** After the conversation, the phone vibrates and displays a notification. A new task, “Complete lab report (due tomorrow),” appears in the phone’s task management interface.
- The user must swap this task with an existing one in their four-slot task list, forcing prioritization.
- **Stress Response:** Upon completing the task swap:
 - Sanity decreases from 100 to 50, introducing mild stress.
 - Subtle anxiety effect begins.
 - Background audio fades slightly and ambient sounds become softer.
 - A quiet internal voice plays: “It’s fine... I can handle it.”
- **Recovery:** After a brief period (2 seconds), sanity gradually restores to 100 and environmental effects return to baseline.
- The interaction ring reappears on the remaining two NPCs, indicating they are now available for interaction.

This act implements the first slope in the Kishōtenketsu structure (Ki), where tension appears but does not dominate. Presence is supported by subtle but consistent changes in audio and visual feedback tied directly to the narrative beat, following Slater and Wilbur’s recommendation that environmental cues track user-relevant events [14].

3.3 Act 2: Rising Pressure (Shō: Responsibilities Accumulate)

3.3.1 Narrative Goal

Demonstrate how responsibilities accumulate from multiple roles (student, teaching assistant) and force ongoing prioritization under increasing time pressure.

3.3.2 Story and Core Interaction

- The user approaches one of the two remaining NPCs.
- This NPC is a professor who requests that the user complete TA grading by the end of the day.
- Player responses:
 - **Overcommitment:** “Yeah, I can do it. No worries.” *Internal voice (after interaction):* “I shouldn’t have said yes. There’s no way I can finish all of this tonight.”
 - **Honest overwhelm:** “Tonight is tough for me. Is there any flexibility?” *Professor:* “I understand, but the department needs this tonight. You’ll still need to complete a significant portion of it.”
- Regardless of the chosen response, the professor communicates urgency and the necessity of completing the work.

3.3.3 Environmental Feedback

- **Phone Interaction:** A notification appears: “New Task Added: Grade TA assignments (due tonight).”
- The user must again swap tasks in their four-slot task list.
- **Stress Response:** Upon task swap:
 - Sanity decreases from 100 to 0 over a longer time frame.
 - Vignette effect intensifies slightly.
- The final NPC’s interaction ring becomes visible.

This act demonstrates role pressure and the stacking of affective load. Mateas and Stern’s concept of guided emotional trajectory [11] ensures that, regardless of player choice, the user experiences a heavier task load and stronger environmental stress signals, pushing the narrative upward. Presence is reinforced by tying environmental effects directly to the player’s task management actions.

3.4 Act 3: Overload (Ten: Breaking Point)

3.4.1 Narrative Goal

Externalize the collapse of cognitive organization and represent the moment when coping mechanisms fail and stress becomes overwhelming.

3.4.2 Story and Core Interaction

- Whisper-like cues externalize self-imposed pressure: “You are late.” “Hurry.” “No time.” “Just finish it.”

3.4.3 Environmental Feedback

- **Stress Escalation:**
 - Sanity drops from 100 to 0 over several seconds.
 - Vignette tightens dramatically, reducing peripheral vision.
 - Chromatic aberration and film grain effects intensify.
 - Heartbeat audio accelerates and increases in volume.
- **Anxiety Audio Sequence:** Random anxiety audio clips play repeatedly (15 seconds total, 1-second intervals), featuring whisper-like intrusive thoughts: “You’re late.” “Hurry.” “No time.” “Just finish it.”
- **Help Paper Appearance:** After 10 seconds of the anxiety sequence, a glowing, pulsing help paper appears in the environment.
- The paper pulses with scale and emission variations to draw attention.
- **User Interaction:** When the user clicks the help paper:
 - The paper animates to the front of the camera.
 - Sanity begins rapidly restoring from 0 to 100.

This act represents the *ten* (twist) in Kishōtenketsu: the environment becomes a direct visualization of intrusive thoughts and cognitive overload, inspired by McMahan’s work on presence through representational strategies that match mental models [12]. The help paper serves as both a narrative device and an agency-restoring interaction that transitions to resolution.

3.5 Act 4: Awareness & Relief (Ketsu: Resolution)

3.5.1 Narrative Goal.

Reframe overload as common, valid, and recoverable while directing users toward mental-health support pathways.

3.5.2 Story and Core Interaction

- A calming voice explains that overwhelm is common among UBCO students and not a personal failure.
- The user views resource poster showing resources on and off campus.

3.5.3 Environmental Feedback

- Vignette, chromatic aberration, and grain effects fade out completely.
- Heartbeat audio fades out completely.
- Camera stabilizes with smooth motion.

The final act transforms stress-induced presence into presence in support. Rather than ending abruptly after overload, the system uses immersive affordances (voiceover pacing and gradual recovery) to direct attention to real-world coping resources, aligning with the project’s goal of mental health awareness and help-seeking.

4 Implemented Systems and Features

This section summarizes the technical architecture of the VR experience, which integrates six major subsystems: the Dialogue System, the Task Management Phone UI, the Notification System, the Sanity/Anxiety System, the Audio Management System, and the Visual Indicator System. These systems operate through an event-driven architecture and a coroutine-based state machine, forming a modular yet interdependent pipeline that drives narrative flow and embodied affective feedback.

4.1 Dialogue System

The dialogue system governs all NPC interactions and player choices. Its core components include `ChoiceUIController`, `DialogueSoundManager`, `NPCConversationData` (ScriptableObject), `NPCInteraction`, and a suite of UI Toolkit (UXML/USS) files that render dialogue panels in world space.



Figure 2: Dialogue UI positioned to the left of an NPC, showing greeting text and two player choice options in world space.

Interaction Flow. When the player enters a $\sim 3\text{m}$ radius around an NPC, `NPCInteraction` detects proximity and displays a glowing yellow interaction ring at the NPC’s feet. Upon clicking, the system triggers a greeting phase: audio plays while text appears in a world-space dialogue panel. After the greeting audio completes (measured through `AudioClip.length`), the system presents two player choices with corresponding audio clips. Selecting a choice triggers the associated voice line, fades out the unselected option, and advances the storyline through a callback to `MainGameScript`.

World-Space Positioning. Dialogue UI is positioned relative to the NPC using offsets in world coordinates:

$$\text{targetPosition} = p_{\text{npc}} + r_{\text{npc}} \cdot x + u_{\text{npc}} \cdot y + f_{\text{npc}} \cdot z$$

to position the dialogue bubble to the left of the NPC, ensuring legibility and consistent spatial anchoring. The UI continuously faces the camera using:

$$\text{transform.LookAt}(\text{camera}); \quad \text{transform.Rotate}(0, 180, 0)$$

Three-NPC Sequential Unlocking. The system implements a mandatory interaction sequence: the first NPC must be completed before the remaining two become available. `NPCInteraction` checks `gameManager.CanInteractWithOptionalNPC()` to determine availability. The first NPC is marked with `isFirstNPC = true`, and upon completion, `MainGameScript.firstNPCInteracted` is set, unlocking the other NPCs. This design ensures narrative pacing and prevents overwhelming the player with multiple simultaneous interactions.

Movement Locking. Player locomotion is controlled by enabling/disabling the `ContinuousMoveProvider` component:

```
void LockPlayerMovement(bool lock) {
    moveProvider.enabled = !lock;
}
```

Movement is locked during:

- Intro sequence (until audio completes)
- All NPC conversations (from greeting to choice completion)
- Ending sequence (during final audio playback)

4.2 Task Management Phone UI

The phone UI acts as the player’s task management device, implemented using `TaskPhoneManager` and `PhoneUIPositioner`. It displays a four-slot “My Tasks” grid and a two-slot “New Tasks” section.

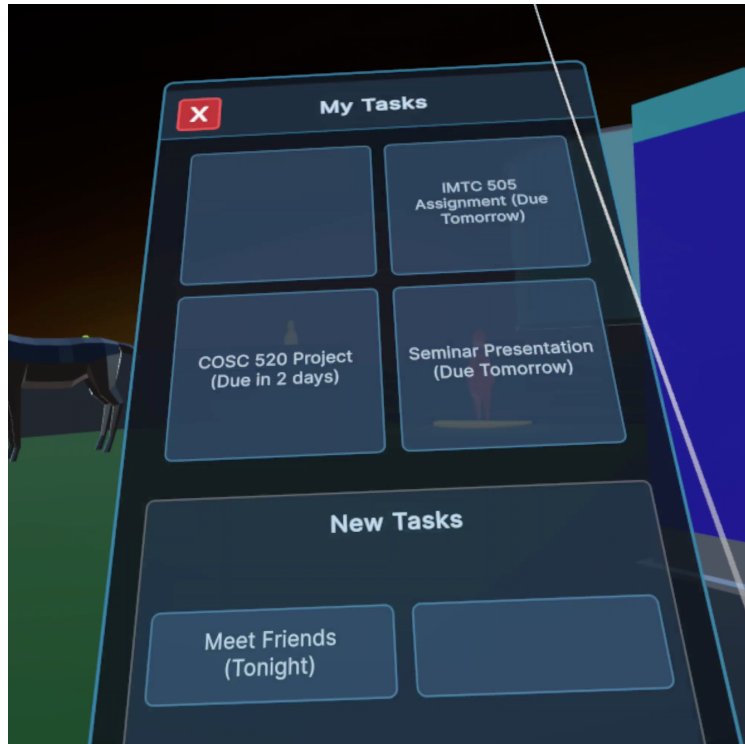


Figure 3: Task management phone interface showing the four-slot task grid with swappable tasks and exit button.

Two-Click Swap Mechanism. Players interact using ray-based XR selection:

1. First click: a task enters a selected state (highlighted via the `.task-selected` USS class with blue border and increased scale).

2. Second click (on different task): the selected task swaps labels with the clicked task.
3. Second click (on same task): deselects without swapping.

The system records which task was replaced:

$$\text{swappedOutTask} = \begin{cases} \text{selectedTask}, & \text{if newTask} = \text{clickedTask} \\ \text{clickedTask}, & \text{if newTask} = \text{selectedTask} \\ \text{null}, & \text{if no swap occurred} \end{cases}$$

An event callback

`OnPhoneClosed(newTask, swappedOutTask)`

is fired to `MainGameScript` when the exit button is pressed, enabling the game controller to progress the narrative and trigger anxiety sequences.

Positioning and Ergonomics. The phone is positioned 1.5m in front of the camera, tilted 30° on the X-axis for comfortable viewing in VR. When inactive (`root.style.display = DisplayStyle.None`), the phone UI is hidden via CSS rather than `GameObject` deactivation, preventing issues with component initialization. The phone remains at a fixed position 10m below the camera when hidden to avoid raycast obstruction.

4.3 Notification System

Notifications are instantiated using `NotificationClick`, positioned using `VRUIPositioner`, and displayed at a fixed offset relative to the camera (0.45m right, 2m forward, at the upper-right periphery of view). The `notificationPos` script ensures the notification follows the camera's base rotation while ignoring jitter from camera shake effects.

Each notification:

1. Activates via `gameObject.SetActive(true)`,
2. Plays a notification sound through `UISoundManager.PlayNotificationSound()`,
3. Repositions immediately via `uiPositioner.ForceReposition()`,
4. Opens the phone UI when clicked via `phoneManager.ShowPhoneWithTask(taskName)`.

Re-Registration Pattern. Because notification objects are repeatedly activated and deactivated, button click events are registered in `OnEnable()` rather than `Start()`, ensuring reliability across multiple notification cycles:

```
void OnEnable() {
    button.clicked -= OnButtonClicked;
    button.clicked += OnButtonClicked;
}
```

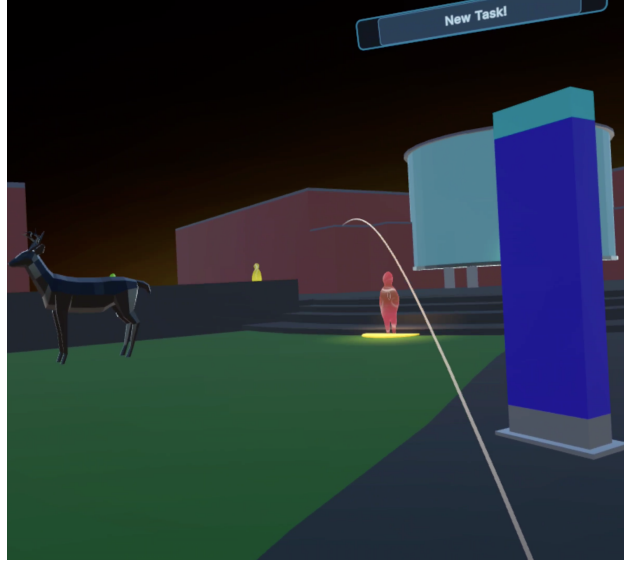


Figure 4: Notification UI positioned in the upper-right periphery of the player’s view, alerting them to a new task.

4.4 Sanity / Anxiety System

The Sanity/Anxiety System operationalizes psychological stress as a continuous scalar variable. The `SanitySystem` maintains a sanity value in $[0, 100]$ and converts it to stress level:

$$\text{stress} = 1 - \frac{\text{currentSanity}}{\text{maxSanity}}$$

where $\text{stress} = 0$ represents calm ($\text{sanity} = 100$) and $\text{stress} = 1$ represents maximum anxiety ($\text{sanity} = 0$).

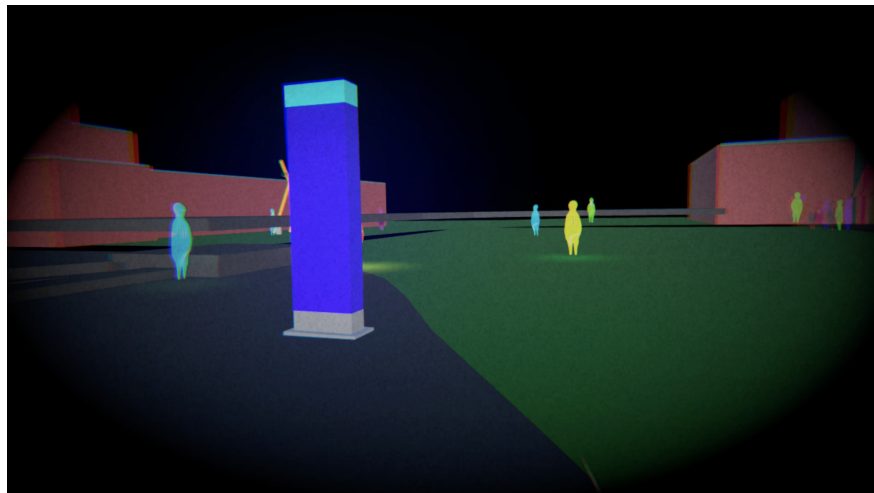


Figure 5: Maximum stress visual effects showing vignette darkening, chromatic aberration, film grain, and reduced visibility at $\text{sanity} = 0$.

The `AnxietyEffectController` subscribes to `SanitySystem.OnStressLevelChanged` events and maps stress values to Universal Render Pipeline (URP) post-processing effects and audio parameters:

- **Vignette:** `Lerp(0, 0.55, stress)` with pulsing overlay
- **Chromatic aberration:** `Lerp(0, 1.0, stress)`
- **Film grain:** `Lerp(0, 1.0, stress)`
- **Camera shake:** `PerlinNoise(t) · stress · maxAngle`
- **Heartbeat volume:** `Lerp(0, maxVol, stress)`

Pulsing Heartbeat System. A sinusoidal pulse synchronized with stress level produces rhythmic vignette oscillations simulating a heartbeat:

$$\text{pulse} = \frac{\sin(t \cdot \text{pulseSpeed}) + 1}{2}, \quad \text{pulseSpeed} = \text{basePulseSpeed} + (\text{stress} \cdot \text{maxPulseSpeedMultiplier})$$

The vignette intensity becomes:

$$\text{vignetteIntensity} = \text{targetIntensity} + (\text{pulse} \cdot \text{pulseMagnitude} \cdot \text{stress})$$

where `pulseMagnitude` scales with stress (up to 0.2), causing the screen edges to darken and brighten rhythmically at rates proportional to anxiety level.

Camera Shake Implementation. Smooth camera shake uses Perlin noise for natural motion:

$$\begin{aligned} x &= (\text{PerlinNoise}(t \cdot 10, 0) - 0.5) \cdot \text{stress} \cdot \text{maxAngle} \\ y &= (\text{PerlinNoise}(0, t \cdot 10) - 0.5) \cdot \text{stress} \cdot \text{maxAngle} \\ z &= (\text{PerlinNoise}(t \cdot 10, t \cdot 10) - 0.5) \cdot \text{stress} \cdot \text{maxAngle} \cdot 0.5 \end{aligned}$$

The resulting rotation is applied to a camera container transform (not the HMD itself) to avoid motion sickness.

Performance Optimization. When $\text{stress} \leq 0.01$, all post-processing effects are deactivated (`effect.active = false`) and values reset to zero, reducing GPU overhead during calm periods. Effects only activate when stress is detectable.

4.5 Audio Management

Audio is managed through two singleton systems: `UISoundManager` and `DialogueSoundManager`.

UISoundManager Channels. `UISoundManager` controls three `AudioSource` components:

1. **UI Audio Source:** One-shot sounds (notification beeps, phone open/close sounds)
2. **Background Audio Source:** Looping ambient music with dynamic volume
3. **Custom Clip Playback:** Narration, anxiety clips, and ending audio sequences

Dynamic Volume Fading. Background music volume adapts to narrative state through coroutine-based lerping:

$$\text{volume}(t) = \text{Lerp}(\text{startVol}, \text{targetVol}, \frac{t}{\text{duration}})$$

Target volumes are:

Normal = 0.3
Anxious = 0.05
Ending = 0.0

DialogueSoundManager. This singleton provides dialogue audio playback with duration tracking:

```
float PlayDialogueClip(AudioClip clip) {  
    if (clip != null) {  
        audioSource.clip = clip;  
        audioSource.Play();  
        return clip.length;  
    }  
    return defaultWaitTime; // 5s fallback  
}
```

This duration value synchronizes UI state transitions with audio completion, ensuring dialogue panels and choices appear only after audio finishes.

Ending Audio Sequence. The ending plays two audio clips back-to-back with a 3-second gap:

```
PlayEndingAudioSequence() -> wait -> PlayEndingAudio2()
```

Background music fades to silence (0 volume) before the sequence begins, then remains muted throughout the ending for narrative impact.

4.6 Visual Interaction Indicators

Two types of visual indicators guide player attention toward interactable objects: interaction rings for NPCs and pulse effects for critical objects.

NPC Interaction Rings. Each NPC has an `InteractionRingController` that manages a cylindrical mesh positioned at the NPC's feet. The ring:

- Uses a yellow emissive material with URP emission enabled
- Rotates continuously: `transform.Rotate(Vector3.up, rotationSpeed * Time.deltaTime)`
- Pulses in scale: $\text{scale}(t) = \text{baseScale} \cdot (1 + \sin(t \cdot \text{pulseSpeed}) \cdot \text{pulseAmount})$

- Pulses in emission: $\text{emission}(t) = \text{baseColor} \cdot \text{intensity} \cdot (1 + \sin(t \cdot \text{pulseSpeed}) \cdot 0.5)$

The ring's visibility is controlled by `NPCInteraction` based on availability:

- **First NPC:** Ring shown immediately at game start
- **Optional NPCs:** Rings hidden until `firstNPCInteracted = true`
- **All NPCs:** Rings disappear when conversation begins

This provides clear, persistent visual feedback about which NPCs can be interacted with from any distance in the environment.

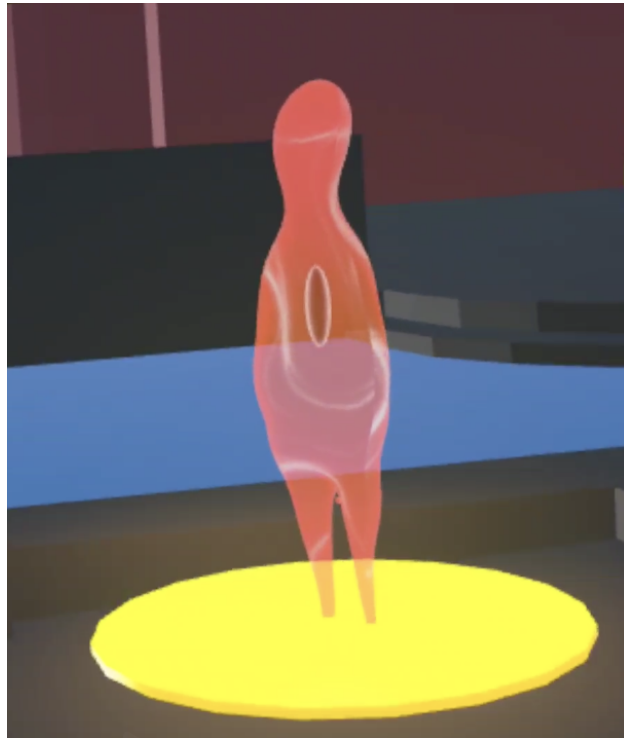


Figure 6: Yellow interaction ring with emissive glow positioned at an NPC's feet, rotating and pulsing to indicate interactability.

Help Paper Pulse Effect. The help paper uses `HelpPaperController` to implement attention-grabbing effects during the overload sequence. When the paper appears (10 seconds into the anxiety phase), it:

- Applies an emissive glow material with white emission
- Pulses in scale: $\text{scale}(t) = \text{originalScale} \cdot (1 + \sin(t \cdot \text{pulseSpeed}) \cdot \text{pulseAmount})$
- Pulses emission intensity: $\text{emissionColor}(t) = \text{Color.white} \cdot \text{intensity} \cdot (1 + \sin(t \cdot \text{pulseSpeed}) \cdot 0.5)$

Default parameters are `pulseSpeed = 2.0` and `pulseAmount = 0.15`, creating a noticeable but not jarring attraction effect. When the player clicks the paper, it:

1. Stops pulsing and glowing
2. Animates to camera position (1.5m forward, tilted 25°)
3. Reverts to original material
4. Triggers `gameManager.OnHelpPaperRead()` to restore sanity

This dual-indicator system (rings for NPCs, pulse for help paper) provides consistent interaction affordances across the experience, reducing cognitive load during high-stress moments when environmental scanning is impaired.

5 Adaptation to Augmented Reality (AR)

Adapting the VR experience to AR requires reinterpreting narrative beats, environmental feedback, and stress-modelling mechanisms within the constraints of real-world perception. Prior work in situated AR storytelling and context-aware overlays [1] [4] demonstrates that emotional narratives can be effectively embedded into real environments when virtual cues supplement rather than replace physical surroundings. The following subsections describe how the VR system would be implemented in AR on the actual UBCO courtyard while maintaining the original narrative arc and affective design principles.

5.1 Spatial Embedding in the Real Courtyard

NPC silhouettes are anchored to precise campus locations using AR plane detection, image targets, or geolocation-based anchors. Modern AR frameworks such as ARCore and ARKit provide persistent world anchors that remain stable across sessions, allowing AR characters to appear consistently near benches, building entrances, or walkways. Their emissive colours (red-orange for stress, blue-green for calm) are rendered as shader-driven glow effects blended onto the camera feed using additive or bloom-based composition.

This approach aligns with research on spatial storytelling and augmented characters that blend abstract visual identities with real-world context [3]. By situating NPCs in familiar campus spaces, the Ki-Shō-Ten-Ketsu arc is reframed through situated interaction:

- “Ki” occurs when students see the first NPC near a common pathway.
- “Shō” unfolds as additional responsibilities appear at different campus points.
- “Ten” emerges when overload cues visually and auditorily saturate the user’s interface.
- “Ketsu” arises when the user reaches locations associated with support resources.

Everyday campus spaces thus become narrative carriers, embodying the “real-world dramaturgy” described by Benford et al. [1].

5.2 AR-Specific Environmental Feedback

Unlike VR, AR must avoid obstructing the user’s physical navigation. Literature on safety in AR [7] emphasises careful modulation of overlays, particularly during emotionally intense content. Thus, overload cues are rendered through lightweight, interface-level augmentation:

- **Semi-transparent notifications** appear as floating UI canvases anchored to the user’s viewport.
- **Peripheral vignetting** simulates narrowing attention without obscuring obstacles.
- **Subtle desaturation** of the camera feed reflects cognitive depletion, implemented through fragment shader passes.
- **Heartbeat and distorted chatter audio** is spatialised through headphones to avoid drawing attention away from environmental hazards.

These cues maintain the affective logic of the VR version while following AR design heuristics for legibility and situational awareness [5].

Implementation Details.

- Screen-space post-processing uses AR Foundation’s camera texture pipeline.
- Notification overload in Act 3 is implemented through repeated instantiation of viewport-anchored canvases driven by time-based coroutines.
- Stress-scalar mapping remains identical to the VR version but is applied to UI opacity, saturation shaders, and audio intensity rather than world-scale effects.

5.3 Contextual Resource Surfacing

Act 4’s VR poster becomes a **location-aware AR support panel**. Situated AR interventions are supported by prior HCI research showing that spatial context enhances perceived relevance and promotes help-seeking [8]. Implementation on the real UBCO campus may use:

- **Image targets:** e.g., scanning a Wellness Centre sign triggers the resource panel.
- **Geofencing:** GPS or Wi-Fi signals auto-spawn the panel when entering a support zone.
- **ARCloud anchors:** persistent shared anchors maintain accurate placement across sessions.

The resource panel may include:

- real-time links to wellness services,
- touch-enabled calling or appointment booking,

- a guided breathing overlay animated in screen space.

These adaptations preserve the design principle that immersive experiences should not merely reproduce stress, but also guide users toward tangible support through environmentally situated cues.

6 Impact and Future Work

6.1 Impact

The system has the potential to support student wellbeing by offering an accessible way to reflect on academic pressure through an experiential medium rather than textual or instructional formats. Its integration of subtle affective cues and resource-oriented guidance positions it as a complementary tool for campus wellness services. Beyond individual use, the experience could inform broader discussions on stress awareness in academic environments, providing institutions with a structured, repeatable intervention that frames stress as recognizable and manageable.

A future deployment of the experience, such as integration into orientation programming or wellness workshops could serve as a reflective tool for students navigating academic demands. Aggregated, anonymized interaction patterns may also offer institutions insights into common stress points or behavioral tendencies, contributing to broader discussions on student well-being and support design.

6.2 Future Work

Several extensions can strengthen the system’s capabilities. Adaptive stress progression informed by physiological measures (e.g., EDA or HRV) would allow the experience to adjust pacing and intensity in real time. The task management system could be expanded to reflect diverse student schedules or coping behaviours. NPC interactions may be enhanced with more context-sensitive dialogue while retaining abstraction. Cross-device support (VR headsets, AR-capable phones, and spatial computing platforms) would broaden accessibility.

There is potential for longitudinal studies examining whether immersive stress visualisation helps influence-seeking behavior or students’ understanding of stress. Controlled experiments could evaluate how abstract NPC design, colour cues, and environmental modulation shape emotional interpretation and presence. The AR adaptation also enables research into situated wellbeing interventions, exploring how location-triggered cues affect engagement with campus support resources. Co-design with students and counselors could refine both narrative content and intervention strategies.

The project contributes to growing interest in using immersive technologies to support mental-health awareness and emotional literacy. By combining narrative structure, environmental feedback, and resource-oriented resolution, Courtyard Reflections offers a model for experiential wellbeing tools that integrate into everyday academic life. This work suggests a pathway for institutions to use VR and AR not only as training or entertainment technologies but as meaningful components of student support ecosystems.

7 Conclusion

Courtyard Reflections demonstrates how immersive technology can be used to meaningfully represent academic stress and guide students toward awareness and support. Through its narrative structure, affective cues, and adaptive interaction design, the project offers an impactful proof of concept of how VR can complement well-being initiatives by externalizing internal struggles in educational environments. The work establishes a foundation for future extensions and research, showing that immersive systems can serve not only as storytelling tools but as practical interventions that support student understanding, reflection, and access to resources.

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The complete source code, technical documentation, and project assets are available in the public GitHub repository at:

https://github.com/aaravg31/Final_project_cosc519

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